

Lab Assignment: Clustering Urban Mobility Patterns Using Bike-Sharing Data

Objective

The objective of this assignment is to:

- Perform data preprocessing and feature engineering for mobility-pattern analysis.
- Apply and compare different clustering algorithms, excluding link-based clustering methods.
- Evaluate cluster quality and stability.
- Interpret the discovered clusters in the context of urban transport planning.
- Demonstrate theoretical understanding of clustering techniques.

Dataset

Bike Sharing Dataset – UCI Machine Learning Repository

URL: <https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset>

Use the **hourly dataset (hour.csv)**, which contains hourly bike rental records from the Capital Bikeshare system in Washington, D.C.

Important attributes include:

- **dteday**: Date
- **hr**: Hour of the day
- **season**: Season
- **yr**: Year
- **mnth**: Month
- **holiday**: Whether the day is a public holiday
- **weekday**: Day of the week
- **workingday**: Whether the day is a working day
- **weathersit**: Weather condition
- **temp**: Normalized temperature
- **atemp**: Normalized apparent temperature
- **hum**: Normalized humidity
- **windspeed**: Normalized wind speed
- **casual**: Number of casual users

- **registered**: Number of registered users
- **cnt**: Total number of rentals

The clustering unit for this assignment is a **day**, not an individual hourly record.

Part 1: Data Preparation and Feature Engineering

1.1 Inspect and clean the data

- Check for missing values, invalid values and duplicate records.
- Convert the date column into an appropriate date format.
- Verify that each date contains a reasonable number of hourly observations.
- Identify any unusual or incomplete days and explain how they were handled.

1.2 Generate daily mobility features

Aggregate the hourly records by date and construct a daily feature vector.

Include at least the following features:

- Total daily rentals
- Average hourly rentals
- Maximum hourly rentals
- Standard deviation of hourly rentals
- Morning peak share: proportion of rentals between 6:00 a.m. and 9:00 a.m.
- Evening peak share: proportion of rentals between 4:00 p.m. and 7:00 p.m.
- Off-peak rental share
- Casual-user proportion
- Registered-user proportion
- Average temperature
- Average humidity
- Average wind speed

You may introduce additional meaningful features.

Do not include date identifiers or duplicate information directly in the clustering feature set.

1.3 Prepare the clustering data

- Examine the distributions of the generated features.
- Identify highly skewed features and apply suitable transformations where necessary.
- Investigate highly correlated or redundant features.
- Standardize the selected features before clustering.
- Clearly justify any features that were removed.

1.4 Exploratory visualisation

Include suitable visualisations such as:

- Histograms or boxplots of daily mobility features
- Correlation heatmap
- Daily rental pattern plots
- Two-dimensional visualisation using PCA

Briefly describe the main patterns observed before clustering.

Part 2: Apply and Analyse Clustering Methods

Apply the following methods to the daily mobility feature vectors.

2.1 K-Means Clustering

- Use more than one method to investigate a suitable number of clusters.
- At minimum, use the Elbow Method and Silhouette Score.
- Report the final number of clusters selected.
- Report and interpret the cluster centroids in the original feature scale.
- Visualise the clusters using PCA or another suitable dimensionality-reduction method.

Interpret what each cluster represents. Possible examples include commuting-dominated days, leisure-oriented days, low-demand days or high-demand days.

2.2 Hierarchical Clustering using AGNES

Apply agglomerative hierarchical clustering using:

- Single linkage
- Complete linkage
- Average linkage

For each linkage method:

- Produce a dendrogram.
- Investigate suitable cluster cuts.
- Discuss differences between the cluster structures.
- Identify which linkage method gives the most meaningful result.

Do not select the final clustering only by visually inspecting the dendrogram. Support the decision using evaluation results.

2.3 DBSCAN

- Use a suitable nearest-neighbour distance plot to investigate the value of `eps`.
- Justify the selected `min_samples` value.
- Report the number of clusters and noise points.
- Investigate the sensitivity of the results to at least two alternative parameter settings.
- Visualise the resulting clusters and noise observations.

Discuss whether the observations identified as noise represent data errors or genuinely unusual mobility days.

Part 3: Clustering Evaluation and Comparison

Evaluate each meaningful clustering result using at least:

- Silhouette Score
- Davies–Bouldin Index
- Comparison of intra-cluster and inter-cluster distances
- Cluster size distribution

For DBSCAN, clearly state how noise observations were handled when calculating evaluation metrics.

Compare the clustering methods based on:

- Cluster separation
- Cluster compactness
- Sensitivity to parameters
- Balance of cluster sizes
- Interpretability
- Treatment of unusual observations

Select the most suitable clustering result for this dataset and justify the decision. The method with the highest value for a single metric should not automatically be treated as the best method.

Part 4: Cluster Interpretation and Validation

For the selected clustering result:

- Produce a profile of each cluster using the original, non-standardized features.
- Compare weekday, weekend, holiday, season and weather distributions across clusters.
- Give each cluster a meaningful descriptive name.
- Explain the mobility behaviour represented by each cluster.
- Identify at least two practical insights that could support bike availability, station rebalancing, maintenance or demand planning.

The variables such as weekday, holiday and season may be used to interpret the clusters even if they were not included as clustering features.

Part 5: Theoretical Understanding

Include concise answers to the following questions in the report:

1. What assumptions does K-means make about cluster shape, size and distance?
2. How does the choice of linkage criterion affect hierarchical clustering?
3. Why is single linkage particularly vulnerable to the chaining effect?
4. Why can DBSCAN identify noise points while K-means assigns every observation to a cluster?
5. Why was feature scaling necessary for this analysis?
6. How can highly correlated features influence distance-based clustering?
7. What are the advantages and disadvantages of using PCA only for visualisation rather than clustering directly on the PCA components?
8. How would Manhattan distance potentially change the clustering results compared with Euclidean distance?
9. Why should clustering quality not be judged using a single evaluation metric?
10. What is the difference between an unusual but valid mobility day and a data-quality outlier?

Submission Requirements

Submit the following:

1. Python implementation

A well-organised and well-commented Jupyter notebook containing:

- Data cleaning
- Feature engineering
- Exploratory analysis
- Clustering implementations
- Parameter selection
- Evaluation
- Visualisations
- Cluster profiling

The notebook must be executable from beginning to end.

2. Short report

Submit a report in **IEEE Conference Format**, with a maximum length of **four pages**, excluding references.

The report should cover:

- Data preparation and feature engineering
- Exploratory findings
- Clustering methods and parameter selection
- Evaluation and comparison
- Interpretation of the selected clusters
- Practical mobility insights
- Answers to the theoretical questions

Large blocks of source code should not be included in the report.

3. Reproducibility

- State the main Python libraries and versions used.
- Use fixed random seeds where relevant.
- Clearly document any decisions made when cleaning or transforming the data.
- Do not manually remove observations simply to improve clustering scores.